



December 2025

ME2302-1 – CONTROL ENGINEERING
(For ME)

Duration: 3 Hours

Max. Marks: 100

Note:

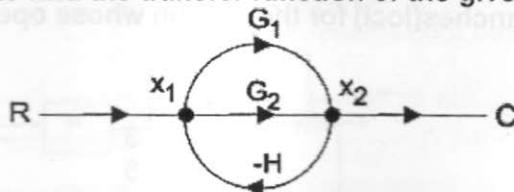
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In closed loop control system with unity feedback, the overall transfer function is
A) $G/(1+GH)$
B) $G/(1+G)$
C) $H/(1+H)$
D) $1/(G+H)$
2. Hydraulic capacitance(C) is
A) dR/dq
B) dh/dt
C) dV/dh
D) dh/dq
3. Force balancing equation of mass element is (when 'x' is displacement)
A) $M(dx/dt)$
B) $M(dx^2/dt^2)$
C) Mx
D) Mx^2
4. Equivalent element for damper in electrical system is
A) Current
B) Capacitance
C) Inductance
D) Resistance
5. Block diagram reduction algebra for parallel block
A) Addition
B) Subtraction
C) Division
D) Multiplication
6. Use mason's gain formula to find the transfer function of the given figure:



- A) $(G_1+G_2)/H$
- B) $(G_1+G_2)/(1-G_1H+G_2H)$
- C) $(G_1+G_2)/(1+G_1H+G_2H)$
- D) $(G_1-G_2)/H$
7. A node having only incoming branches in signal flow graph.
A) Output node
- B) Input node
- C) Summing node
- D) Takeoff
8. In closed loop control system, the overall transfer function is
A) $1/(G+H)$
- B) $G/(1+H)$
- C) $H/(1+G)$
- D) $G/(1+GH)$
9. Larger time constant corresponds to -----
A) Fast response
- B) Slow response
- C) No response
- D) No effect
10. Time constant of first order system is represented by transfer function $\frac{C(S)}{R(S)} = \frac{4}{s+4}$.
A) 4 sec
- B) 2 sec
- C) 0.5 sec
- D) 0.25 sec

11. Mathematically Step input is represented as
 A) $r(t) = A$, when $t \geq 0$
 B) $r(t) = At$, when $t \geq 0$
 C) $r(t) = \frac{At^2}{2}$, when $t \geq 0$
 D) $r(t) = A \sin \omega t$, when $t \geq 0$
12. Peak overshoot when $\omega_n = 4 \text{ r/s}$, $\xi = 0.5$
 A) 16.3%
 B) 10.3%
 C) 23.3%
 D) 1.63%
13. Non repeated roots on the imaginary axis makes the system
 A) Stable
 B) Unstable
 C) Marginally stable
 D) No effect
14. Routh Hurwitz criterion gives
 A) Number of roots in the right half of the s-plane
 B) Value of the roots
 C) Number of roots in the left half of the s-plane
 D) Number of roots in the top half of the s-plane
15. If the gain margin and Phase margin are positive, then system is
 A) Unstable
 B) Stable
 C) Limitedly stable
 D) No effect
16. In Nyquist criterion for stability, if $Z=0$.
 A) $N=P+1$
 B) $N=P-1$
 C) $N=P$
 D) $N=0$
17. Centroid point of asymptotes can be obtained by
 A) $(\sum P - \sum Z)/(P-Z)$
 B) $(\sum P - \sum Z)/(P+Z)$
 C) $(\sum Z - \sum P)/(P-Z)$
 D) $(\sum P - \sum Z)/(Z-P)$
18. $m(t) = K_p e(t) + K_i \int e(t) dt + K_d \frac{de(t)}{dt}$, it represents control action
 A) Proportional and Integral
 B) Proportional and derivative
 C) Integral and derivative
 D) Proportional plus Integral plus derivative
19. Correct statement with respect to root locus plot procedure
 A) The root locus can exist on the real axis to the left of an odd number of real poles or zeroes.
 B) The root locus can exist on the real axis to the right of an odd number of real poles or zeroes.
 C) The root locus can exist on the real axis to the left of an odd number of real poles only
 D) The root locus can exist on the real axis to the left of an odd number of zeroes only
20. Number of root locus branches(loci) for the system whose open loop transfer function is given by $G(S)H(S) = \frac{K}{s(s^2+2s+2)}$
 A) 2
 B) 3
 C) 4
 D) 5

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Explain the open and closed loop control system with block diagram and example.
 b) Find the overall transfer function of the system shown in the fig.1(b) using Reduction algebra

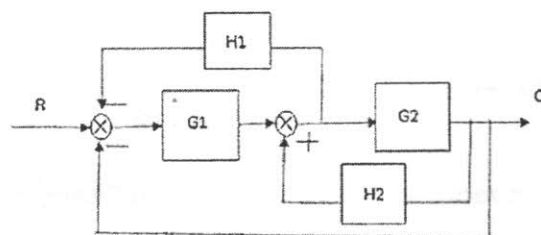


Figure.1(b)

Marks	BT*	CO*	PO*
8	L2	1	1

8	L3	2	2
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2. a) Obtain the transfer function $Y(s)/F(s)$ of mechanical system as shown in the fig.2(a)

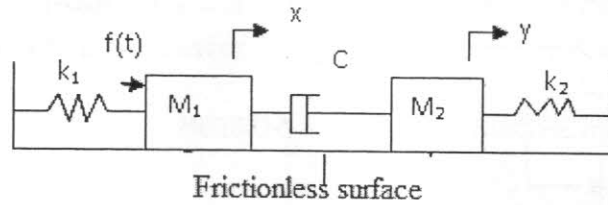


Figure.2(a)

8 L3 1 2

- b) Find the overall transfer function of the system shown in the fig.2(b) using (i) Mason's gain formula and (ii) Block diagram reduction

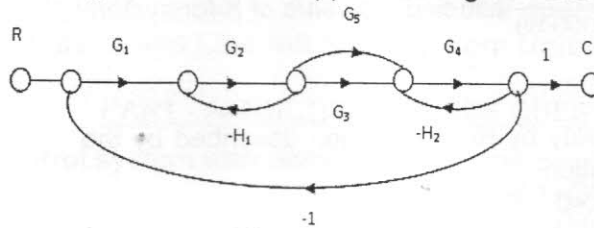


Figure.2(b)

8 L3 2 2

3. a) Find the transfer function for the electrical network shown in fig.3(a)

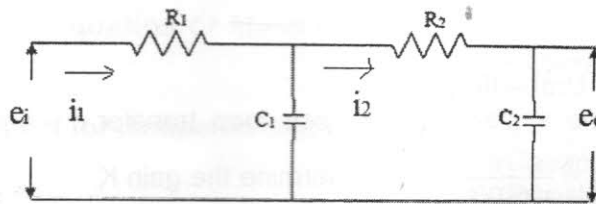


Figure.3(a)

8 L3 1 2

- b) Determine transfer function for a system whose block diagram is shown in figure.3(b) using reduction algebra

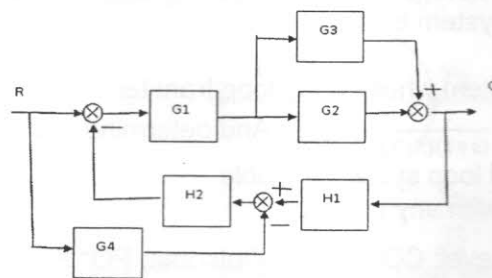


Figure.3(b)

8 L3 2 2

Unit - II

4. a) Derive the response expression of 2nd order under damping system for unit step input.
- b) Draw the Nyquist plot and determine the stability for the open loop transfer function of the system $G(S)H(S) = \frac{1}{s(1+2s)(1+s)}$

8 L3 3 2

8 L4 4 2

5. a) For the feedback system as shown in fig.5(a), find K_1 & K_2 So that peak overshoot is 1.6% and corresponding peak time is 1.2 sec.

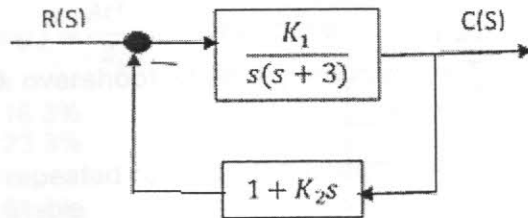


Fig.5(a)

- | | | | | |
|---|---|----|---|---|
| b) Sketch the Bode plot of a system whose open loop transfer function is given by $G(S)H(S) = \frac{K}{s(s+1)(s+10)}$ and find the value of K for system to be stable. | 8 | L3 | 3 | 2 |
| 6. a) Find the value of K for stability by Routh criterion described by the following characteristic equation
(i) $F(s) = s^4 + 4s^3 + 13s^2 + 36s + K = 0$
(ii) $F(s) = s^3 + 5s^2 + (K-6)s + K = 0$ | 8 | L3 | 3 | 2 |
| b) Sketch the Nyquist plot for the system whose open loop transfer function given by $G(S)H(S) = \frac{K(1+s)}{(1-s)}$ Determine the value of 'K' for which the system is stable. | 8 | L4 | 4 | 2 |

Unit – III

- | | | | | |
|---|----|----|---|---|
| 7. a) Sketch the root locus for the system whose open loop transfer function is given by $G(S)H(S) = \frac{K}{s(s+3)(s+5)}$ And determine the gain K which will give the closed loop system is stable. | 8 | L4 | 5 | 2 |
| b) Sketch the root locus for the system whose open loop transfer function is given by $G(S)H(S) = \frac{K(s+6)}{s(s+1)(s+2)}$ And determine the gain K which will give the closed loop system is stable. | 8 | L4 | 5 | 2 |
| 8. a) Sketch the root locus for the system whose open loop transfer function is given by $G(S)H(S) = \frac{K}{(s+1)(s+3)(s^2+2s+2)}$ And determine the gain K which will give the closed loop system is stable. | 12 | L4 | 5 | 2 |
| b) What is control action? Explain with any one type. | 4 | L2 | 5 | 1 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
